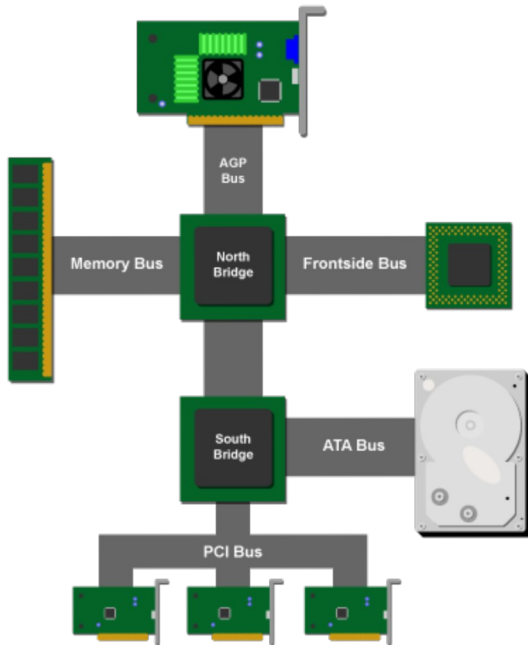


Computer System Fundamentals

by
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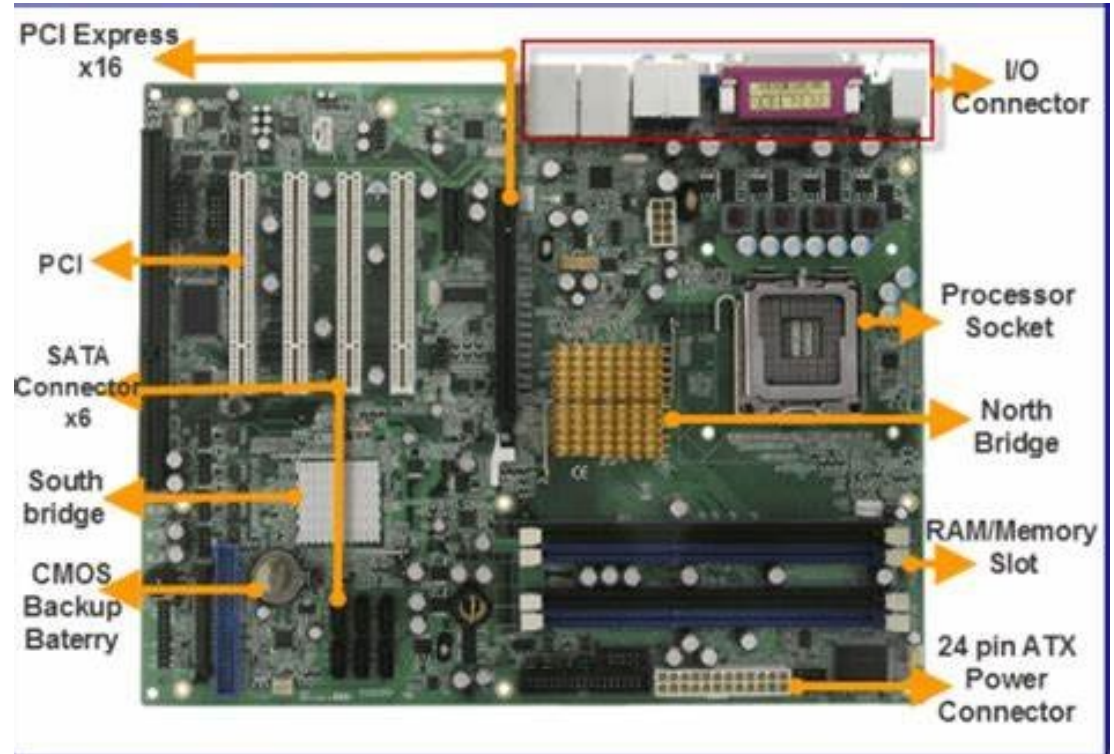
Computer System (Motherboard)

- Sockets/Slots : CPU Socket, Memory Slots
 - System Bus
 - North Bridge (Memory Controller Hub)
 - Frontside Bus (CPUs.)
 - AGS Bus (GPUs)
 - Memory Bus



- System Bus
 - South Bridge (IO Controller Hub)
 - PCI (parallel, shared bus)
 - PCI-e (serial, point-to-point)
 - SATA (serial) – drives, SSDs

- Ports (Connectors)
- Power connector
- CMOS Battery
- ROM chips (BIOS)
- Clocks (RTC, CPU)



What is System Bus ?

A bus is a collection of wires connecting components (CPU, memory, disks, etc.) of a motherboard.

Address Bus

- Connects Physical memory addresses to CPU and IO devices
- Determines the maximum size of memory
- Bus lines:
- 32-bit address $\Rightarrow 2^{32}$ (4 Gigabytes)
- 64-bit address $\Rightarrow 2^{64}$ (16 Exabytes)
- 48-bit address $\Rightarrow 2^{48}$ (256 Terabytes)
- **The 64 bit processors do not use the full 64 bit lines (to save transistors). Most use 48 bits or less.**

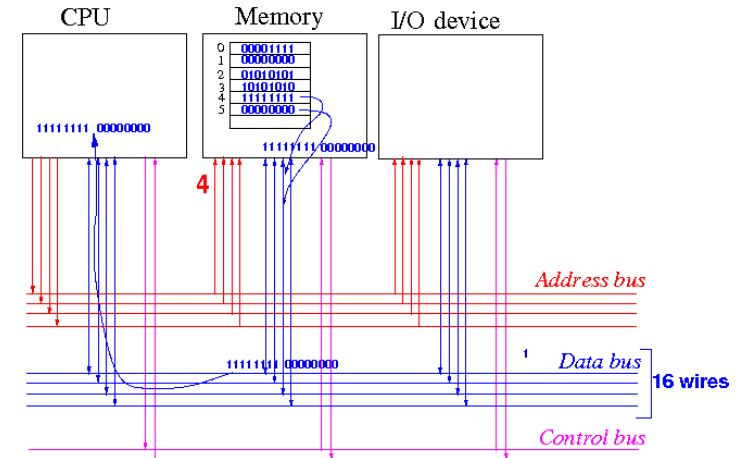
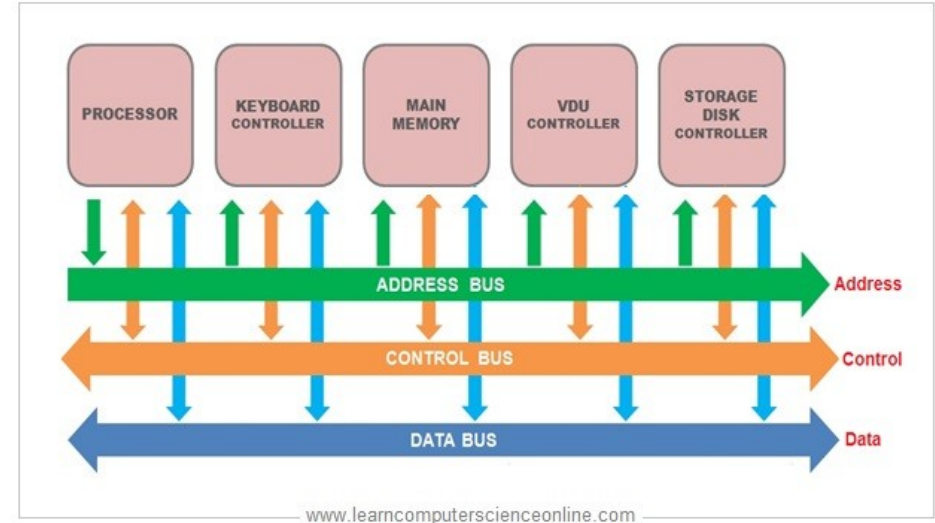
Control Bus

- Used to control data exchange flow (read, write, request access, grant access, acknowledge, etc.)
- Control bus size can vary from 16-bit to 32-bit or more.

Data Bus

- Used to exchange (transmit/receive) data - bidirectional flow
- Standard Data bus size is **32 bits**
- A higher data bus size would mean higher performance (a 64-bit data bus can transfer twice the amount of data compared to a 32-bit data bus)
- Advanced system have **64-bit** to **96-bit** data bus sizes

Motherboard Bus Structure



What is Memory ?

Primary Memory

- **RAM (Random Access Memory) - SRAM/DRAM**
 - When organised as a 2D circuit: address lines translate to a word line and columns to a bit line.

Cache Memory: A reserved area for temporary storage

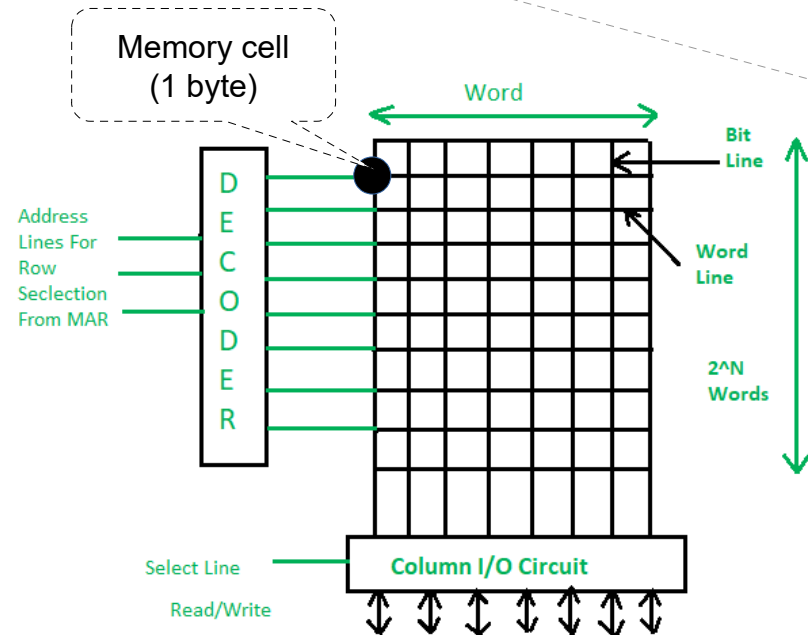
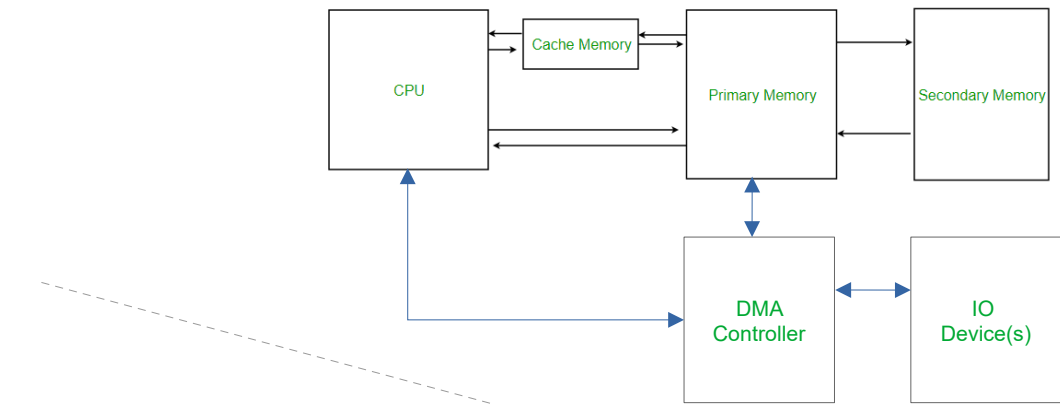
- **CPU Cache**
- Temporary storage within CPU
- **L1 Cache**
 - low capacity but fast,
 - within the CPU core
 - 64K (per core)
- **L2/L3 cache**
 - slow and shared by all CPU cores
 - 6 to 12MB (or more)

Secondary Memory

- Hard disks, SSDs, CDs, DVDs, etc.

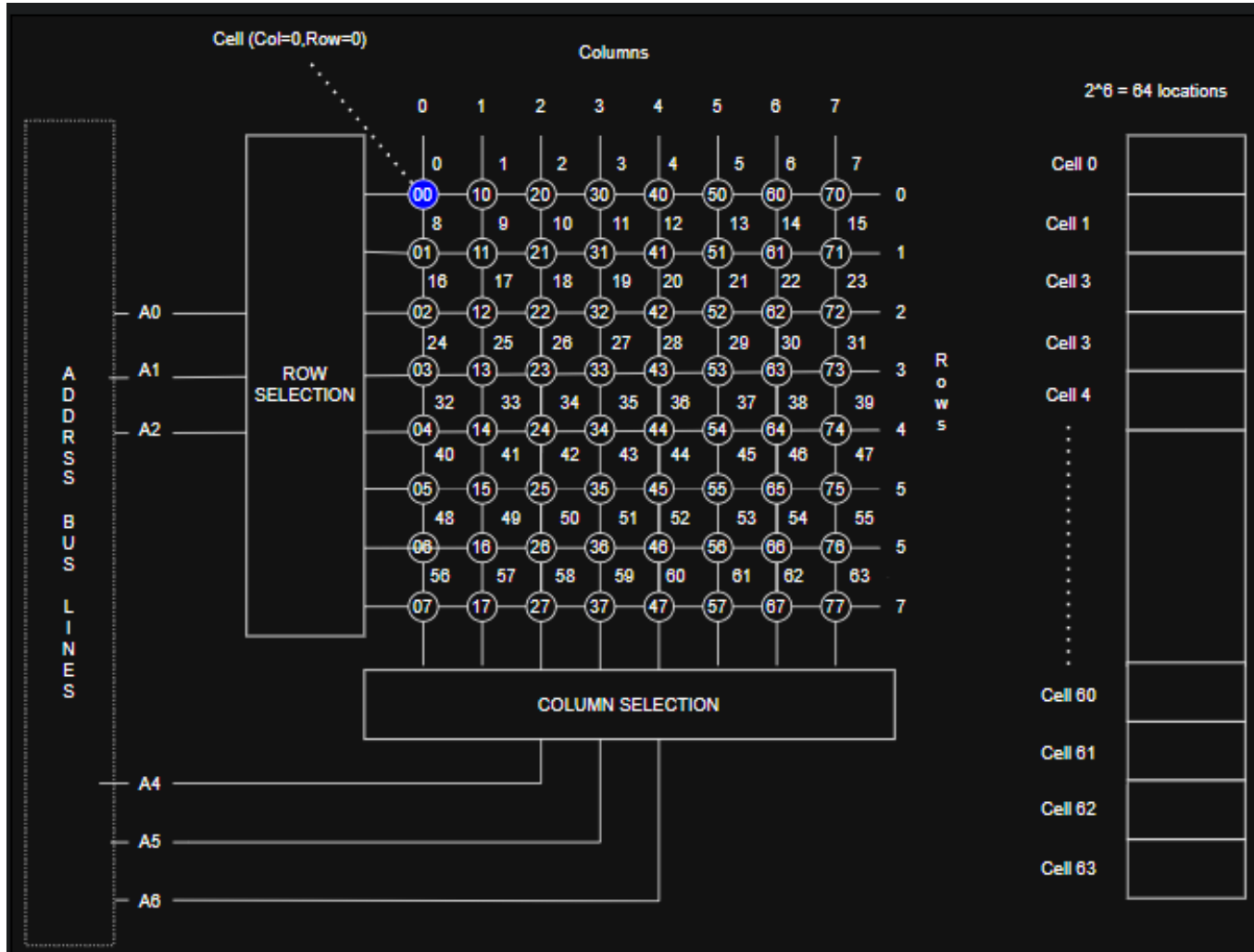
DMA (Direct Memory Access) Controllers

- These are specialized controllers to do IO operations (memory block read/write)
- CPUs delegate IO operations to DMA controllers (for better coordination and performance)



2D Memory Organization

Physical vs Logical Memory Layout (Example: 6 address lines, $2^6 = 64$ locations)



Select Cell 30:

30 => translated to Column=6 Row=3
=> Cell (6, 3)

Select Cell 54:

54 => translated to Column=6 Row=6
=> Cell (6, 6)

Select Cell 17:

17 => translated to Column=1 Row=2
=> Cell (1, 2)

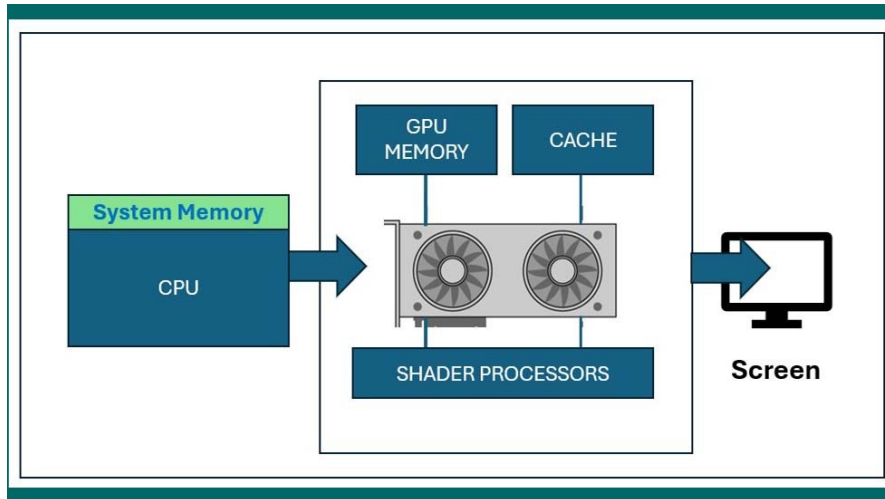
What is a processing controller?

General Purpose CPUs (Central Processing Units)

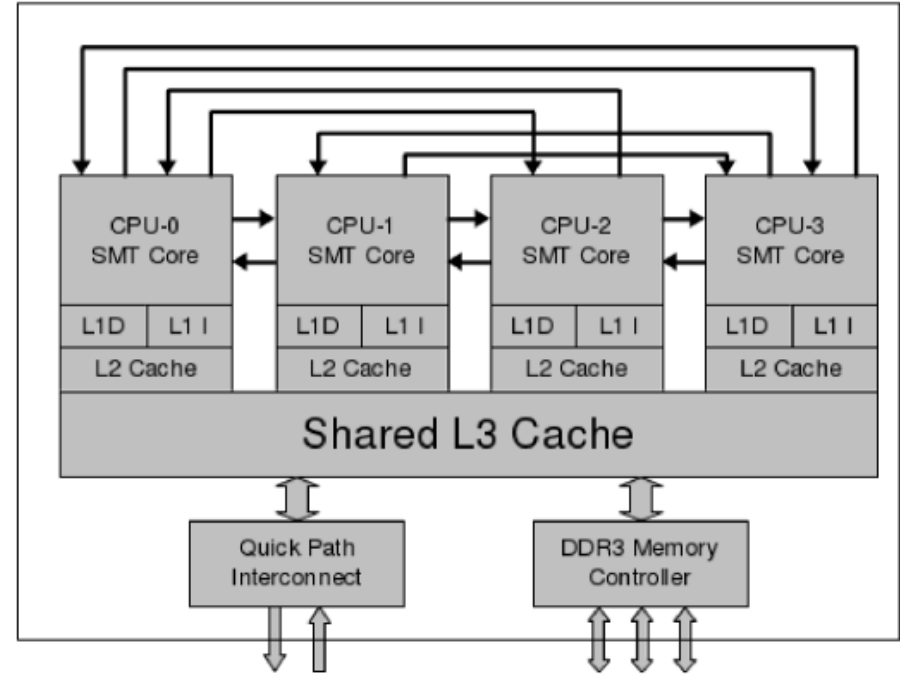
- Intel Core / AMD Ryzen (32-bit / 64-bit architecture)
- FPGA (Field Programmable Gate Arrays)

Special Purpose CPUs:

- DSP (Digital Signal Processing)
- ASIC (Application Specific Integration Chip)
- GPU (Graphics Processing Unit) – e.g. NVIDIA
- TPU (Tensor Processing Unit) - AI dedicated chip
- Microcontrollers (Integrated circuits)



Intel Core CPU Block Diagram



CPU Instruction set:

- Every CPU supports a set of instructions (a collection of commands) that it can execute.
- The instructions are essentially **machine language (sequence of binary digits)** that the CPU can directly interpret.

1Knowledge Check (Computer System)

1) Which of the following buses used by the CPU?

[a] Address [b] Data [c] Control [d] IO Bus

2) Which of the following are general purpose CPUs?

[a] DSP [b] Intel Core [c] AMD Ryzen [d] NVIDA GeForce

3) What is the maximum physical memory with 32-bit address space?

[a] 2^{32} [b] 4 Gigabyte [c] 256 Petabytes [d] 32

4) Which of the following cache is shared by the CPU cores?

[a] L1 [b] L2 [c] L3 [d] System Cache

5) Which of the following does data copy operation?

[a] CPU [b] DMA Controller [c] Hard Disk [d] RAM

6) Which of the following are secondary memory?

[a] RAM [b] Cache [c] ROM [d] Hard Disk

7) Which of the following are primary memories?

[a] RAM [b] ROM [c] DVD [d] Hard Disk

8) On a 64-bit system, the CPU will use all the 64 bit lines.

[a] True [b] False

9) On a 64-bit system, the data bus size must be 64 bit.

[a] True [b] False

10) Modern system use which of the following interfaces?

[a] PCI [b] PCI-e

11) What decides the maximum size of memory on the system?

[a] Data Bus [b] Address Bus [c] Control Bus

12) What is the form of the instruction set of a CPU ?

[a] Machine language [b] 0s and 1s [c] High level commands

Operating System, Processes & Threads

What is a Operating System?

A special software that provides a controlled platform for applications to access the underlying hardware resources.

Example: Windows, Linux, Mac

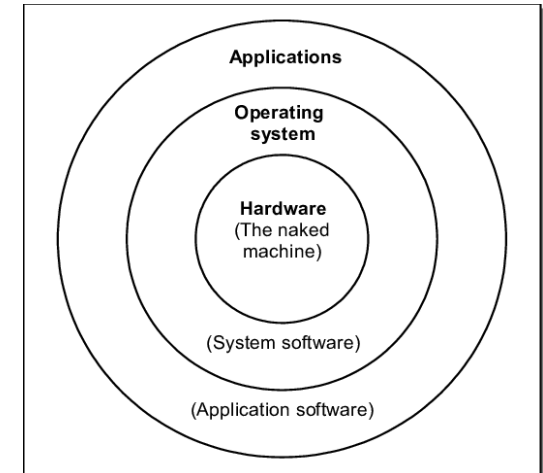
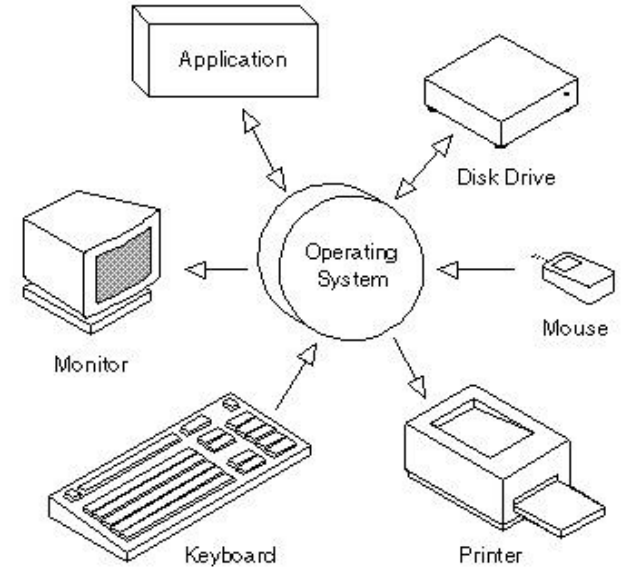
Key features of Operating System:

- Process Management
- Memory Management
- File System (FAT, NTFS, etc.)
- Device Drivers

Process Management

The operating system implements scheduling and access sharing algorithms for the processes. A simple algorithm is **Round Robin** where CPU time is divided between processes (applications) and each process gets a chance to runs its instructions during the available time slot.

Video: [Round Robin Scheduling](#)



Types of processes?

- **Front End**

- These are processes that have a User Interface and require user interaction. e.g, Word, Excel, Browsers, etc
- .
- A front end application have one or more **UI threads** (threads that need user interaction). Generally one UI thread is sufficient.
- A front end application have any number of **back-end threads** also called **Worker threads**.
-

- **Back End**

- These are processes that do not have a user interaction and can run without user interaction. e.g. Services, daemons, etc.
-
- A back-end process can have any number of **back-end threads** also called **Worker threads**.

What is a Thread?

- A process does not run by itself, It needs one or more threads to run.
- A thread can spawn (create) another thread
- Every process has a startup code that creates the main thread that runs the process. e.g. every program has a main or startup function.
- A thread is like a sub-process within process
- A process must have at least one thread
- A process can have any number of threads
- Threads allow concurrency and parallel execution within a process

How memory is allocated to a process?

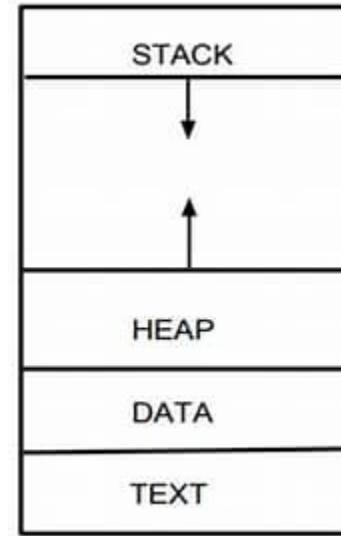
A process is an executable having :

1. code (machine level instructions)
2. data structures & primitive data used by functions
3. data structures created at runtime
4. constant and static data shared by program function

When a process is loaded in memory by the OS, it is allocated following memory sections:

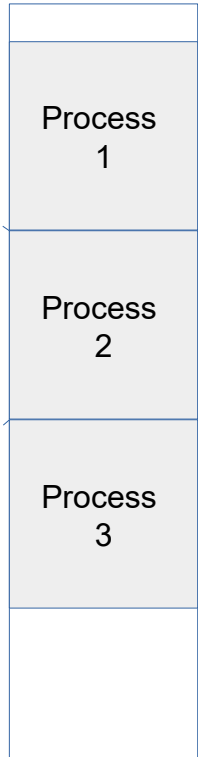
1. **Stack** – This is the memory where all program temporary data such as local variables and function parameters are stored.
2. **Heap** – this is the memory where data created at runtime by explicit allocation. This is dynamic.
3. **Data** - stores global and static variables
4. **Text** or **Code** – the program instructions kept here

Every thread has its own stack.
If there are 5 threads, there will be 5 stacks



Memory allocated for
the process by OS

memor



If too many temporary data is created, Stack memory can overflow and program will crash.

If program tries to access data outside heap memory, it will crash or fail.

Process Memory allocation (Example)

```
#include <stdio.h>

void FunctionA(int a);
void FunctionB(char b);

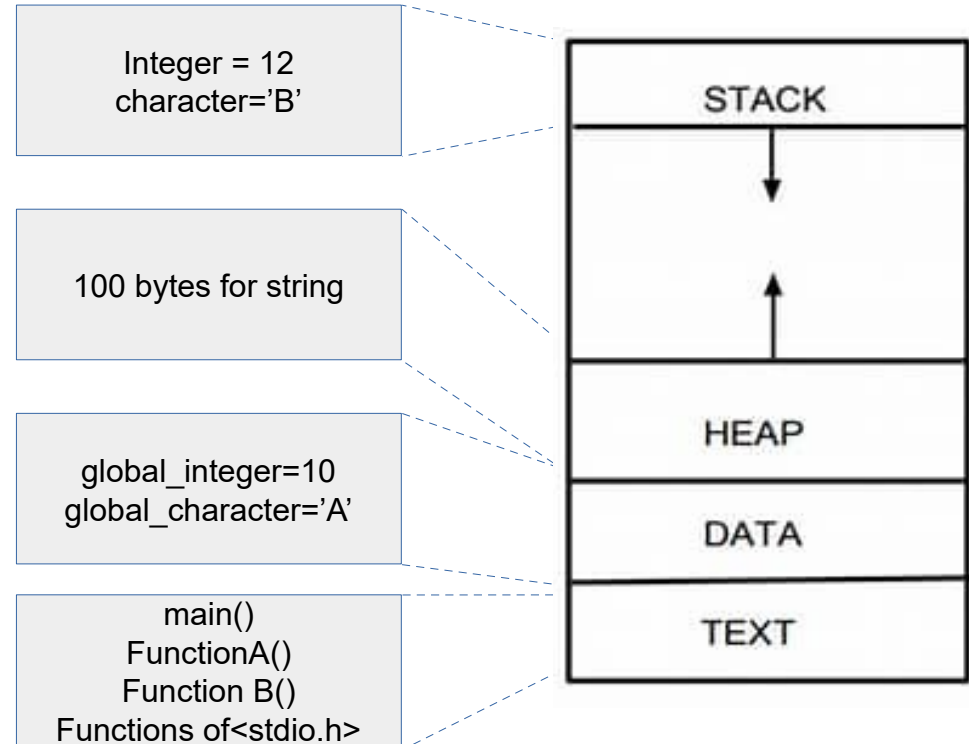
int global_integer = 10;
char global_character = 'A'

int main() {

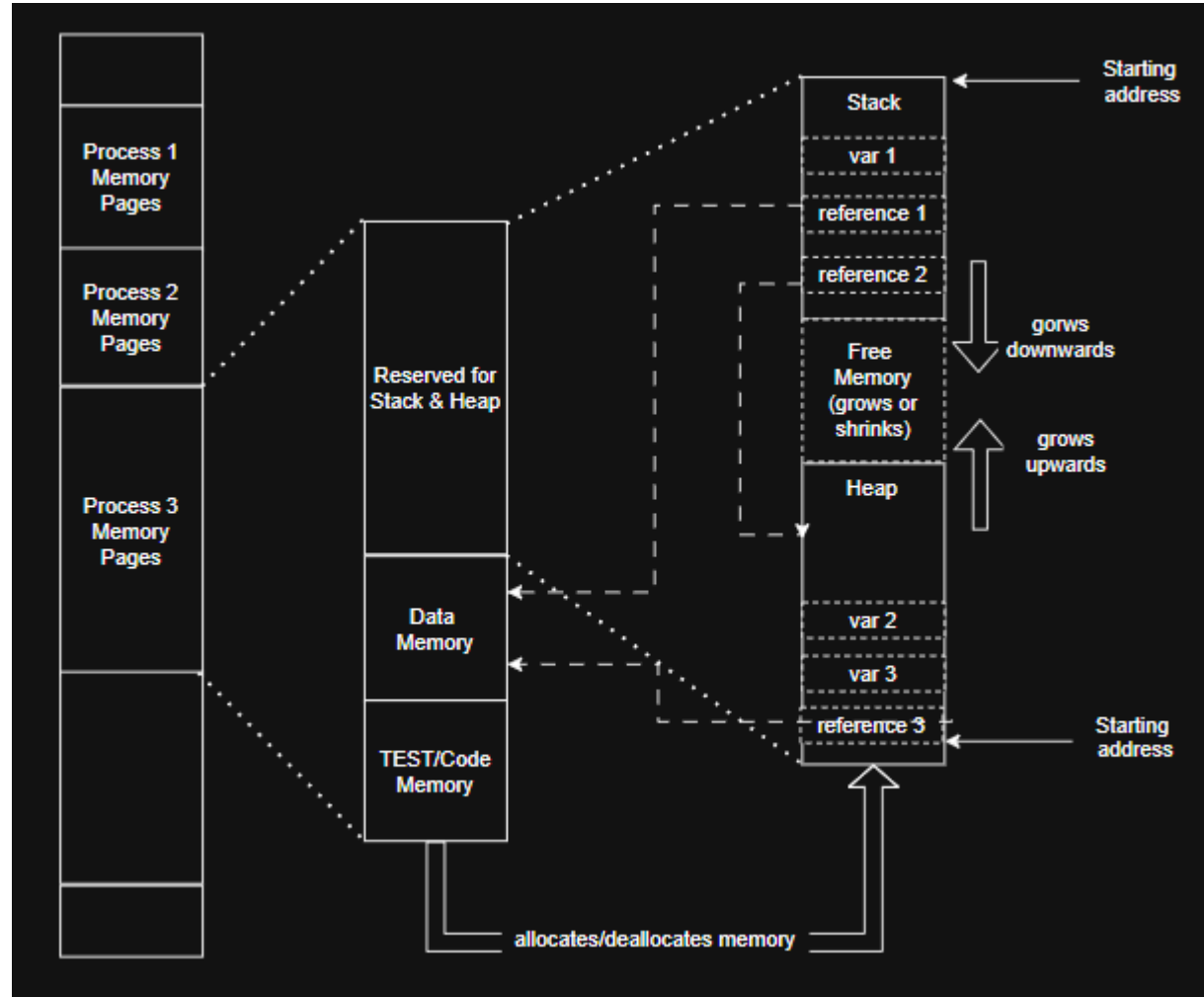
    int integer = 12;
    char character = 'B';
    char* string = (char*)malloc(100);
}

void FunctionA(int a)
{
    printf("Integer value: %d\n", a)
}

void FunctionA(char c)
{
    printf("Integer value: %d\n", a)
}
```



Process Memory allocation



²Knowledge Check (OS, Process & Threads)

1) How many threads are required for a process to run?

[a] 1 [b] 2 [c] 3 [d] 4

2) A thread is a a process within a process.

[a] True [b] False

3) A heap is a memory area where programs's temporary data is stored?

[a] True [b] False

4) A Stack is shared by all threads of a process.

[a] True [b] False

5) Which of the following can be part of a Front End process?

[a] UI Thread [b] Worker Thread [c] Both [d] None

6) A Worker thread is a background thread.

[a] True [b] False

7) The program functions and execution code is stored at:

[a] Stack [b] Heap [c] Data [d] Text

8) Heap memory is shared by all threads of the rocess.

[a] True [b] False

9) The Data section of a process stores:

[a] Instructions [b] Constant variable [c] Global variables

10) A worker thread can interact with user

[a] True [b] False

11) Applications can access the hardware via

[a] Operating System [b] Processes [c] Threads

12) An application has unlimited stack and heap

[a] True [b] False